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SET-I

Answer 1.

(a) Definition of Sustainable Development:

Meeting present needs without compromising future generations' ability to meet theirs.

Three Principles:

1. **Environmental Integrity:** Preserving natural systems.
2. **Economic Efficiency:** Using resources responsibly.
3. **Social Equity:** Fair distribution of opportunities.

(b) Three Pillars of Sustainability:

1. **Environmental Pillar:** Conserving natural resources.
2. **Economic Pillar:** Growth with responsible resource use.
3. **Social Pillar:** Fair access to opportunities and equality.

Answer 2.

(a) Types of Food Chain:

1. **Grazing Food Chain:** Starts with producers (e.g., grass → grasshopper → frog → snake → hawk).
2. **Detritus Food Chain:** Starts with decomposing organic matter (e.g., dead leaves → earthworms → birds).

Answer 3.

(a) Effects of Deforestation:

1. Loss of biodiversity
2. Climate change
3. Soil erosion
4. Water cycle disruption

(b) Renewable and Non-Renewable Resources:

1. **Renewable Natural Resources:** Replenish naturally (e.g., sunlight, wind, forests).
2. **Non-Renewable Natural Resources:** Finite and deplete over time (e.g., fossil fuels, minerals).

SET-II

Answer 4.

(a) Negative Consequences of Excessive Water Resource Exploitation:

1. Groundwater depletion
2. Water pollution
3. Ecological imbalance

(b) Threats to Biodiversity:

1. Habitat loss
2. Climate change
3. Pollution
4. Invasive species

Answer 5.

(a) How Air Pollution Affects Human Health:

- Respiratory issues (e.g., asthma)
- Cardiovascular diseases
- Neurological effects

(b) Impacts of Water Pollution:

1. **Environment:** Aquatic life destruction, ecosystem damage.
2. **Humans:** Waterborne diseases, chronic illnesses.

(c) Effects of Soil Pollution:

1. **Soil Organisms:** Reduced fertility.

2. **Agriculture:** Poor crop yield.
3. **Humans:** Health issues (e.g., cancer).
4. **Ecosystems:** Habitat destruction.
5. **Plants/Animals:** Growth and reproduction harm.

Answer 6.

(a) Harmful Effects of Acid Rain:

1. Soil damage
2. Water body acidification
3. Infrastructure corrosion
4. Forest damage

(b) Population Distribution and Environmental Influence:

1. **Dense Areas:** Resource overuse and pollution.
2. **Sparse Areas:** Underutilized resources.
3. **Global Impact:** Regional deforestation and habitat loss.